

Socio-cultural perspectives to innovative technologies

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Agenda for today

13.15 – 14.00: Session 1

- Introduction of the course syllabus, learning outcomes and assignments (10 min)
- What is a socio-cultural perspective to innovative technologies and design?
  (25 min)
- Introducing ourselves (10 min)

14.00-14.10 Break

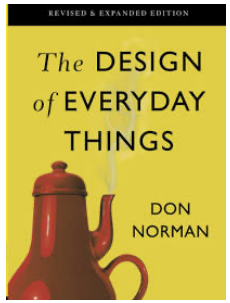
14.10-15.00: Session 2

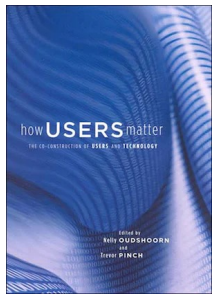
- Why do users matter? Discussion (15 min)
- Social science methods in engineering fields (10 min)
- Examples on user studies (1o min)
- Group formation for final assignment (10 min)

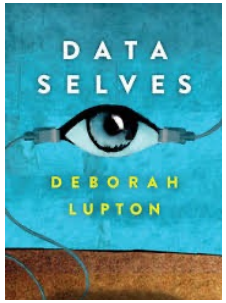


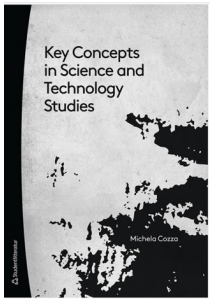
Course syllabus and learning objectives

- Science and Technology studies (STS), Human-Computer Interaction, Design studies and social sciences
- The relationship between technology and society; the interplay between technology and social and cultural factors
- Adoption, use, implementation of new technology in the everyday life and organisational environments
- Users role in the design and development process
- Methods to conduct user studies and understand usage patterns in different socio-cultural contexts
- Development of a study design that involves users in the design process of technology in an innovative manner









Individual assignments

- **5 reaction papers** (1-2 page per paper) after each lecture
- Aim is to summarise, analyse and reflect the content of each lecture and connect the content of your lecture to your own fields of study.
- Participation in seminars and discussions: Short group presentation for each seminar (10-15 min/per group), active participation in the disussion

Group assignment

- **Final assingment**: User study for an innovative health technology

- Describing the **context** and technology
- Introducing the **user group(s)**
- Selecting a suitable **theoretical concepts** describe in the course
- Select and justify one or more relevant **methods** to study users
- **Collect data from the users**

Seminars

- Purpose is to present the *current stage* of the user study (10-15 min per group)
- Give and receive feedback from teacher and other students
- Note: Seminar on 22th April and 6th May:

  - Submit early draft in Canvas

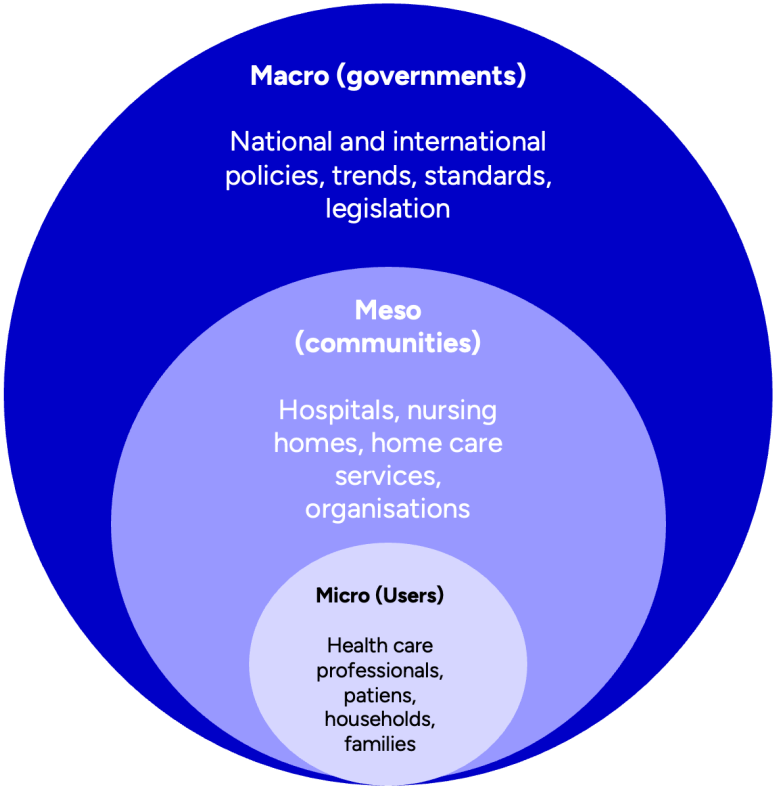
  - Read other students proposals

  - Act as an opponent for another student

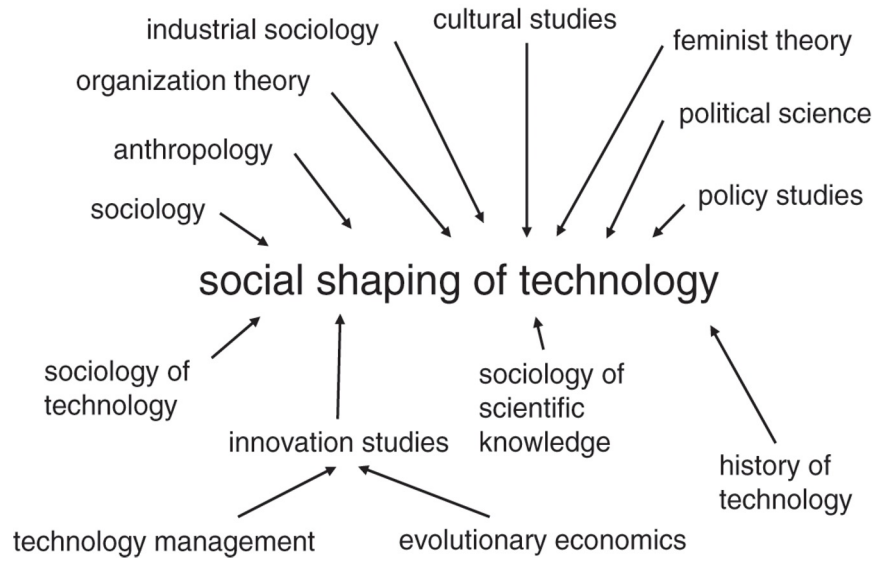


Socio-cultural perspectives to innovative technologies

- Investigating users at different levels and contexts
  (individual - community - culture and society)
- Contextualisation of technology: 'From laboratory in the wild'
- Technology adoption is not a linear process – users shape technology (meaning – purpose – use)
- Technologies are not value neutral - they affect users and shape their identities, practices and values



Socio-cultural perspectives to innovative technologies



Cambridge University Press: Chapter 6 - The Social Shaping of Technology (SST)

- Critique towards technological determinism: Technologies do not develop outside societal, cultural or economical processes
- Shifting the focus from technological development towards technology adoption, use and implementation
- Innovation is as collective processes depending on the societal, cultural, economic influences
- Mutual Shaping: Technology is socially constructed (social constructionism)

Users vs. designers

- **Users' perspective to innovation and design**
- Users sources and adopters of innovation
  - Users modifying technology for other demands
  - Rejection, 'misuse' and 'non-use'
  - Implementation of technology in home

- **Designers' perspective to configuring users (Lecture 3)**
- Designers creating personas and scripts
- Designers configuring needs and implementing stereotypes

- **User involvement and user-centered methodologies** (Lecture 4, 6-8)

The maddening story of why microwaves were first marketed only to men

Welcome to the world of gadgets for guys, also known as 'brown goods'



Gender and technology

- In what way new technologies are gendered – impacted or influenced by gender norms and stereotypes?
- What kind of hidden assumption about the gender roles are encoded in new technologies?
- In what way gender impacts the use of new technologies (e.g. care work, self-care practices)
- In what way new technologies impact the different genders in different ways? (e.g. AI in gendered health)

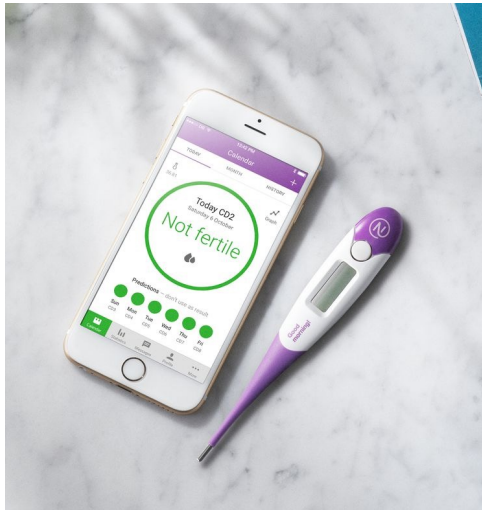


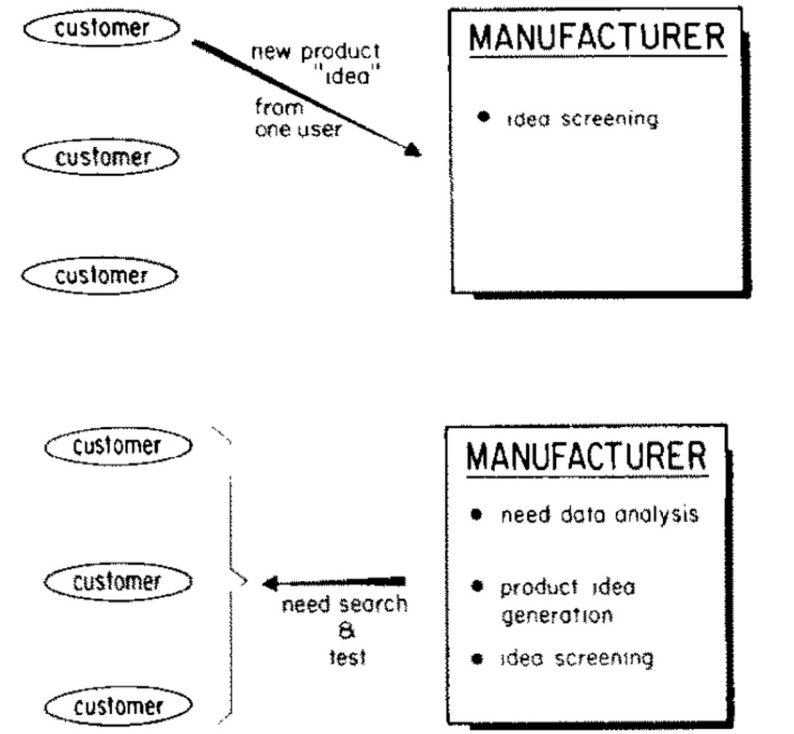




Mutual Shaping

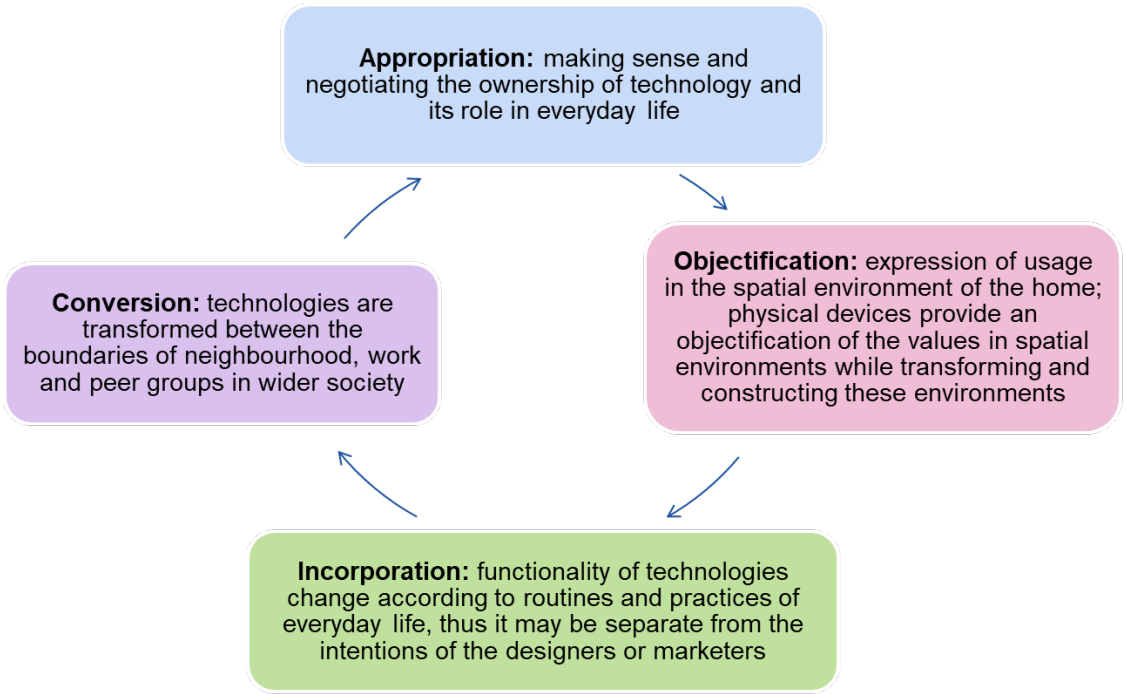






Interests, motivations, expertise, politics, and material objects are closely intertwined

Domestication of technology



- People construct their experience with technologies through social meaning-making processes
- Users and households are engaging with the products and their meanings: possibility to **change the purpose and meaning of technology**

Why do users matter?

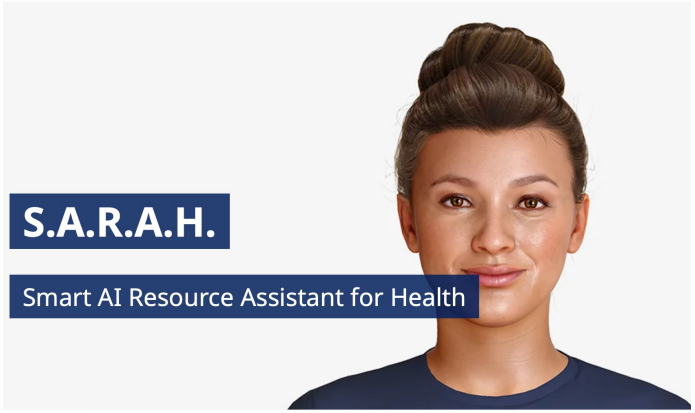
- Discussion in small groups (15 mins): [bit.ly/3nb2FuU](bit.ly/3nb2FuU)

- Choose one example and answer the questions









Why do users matter?

- GoogleDoc: bit.ly/3nb2FuU

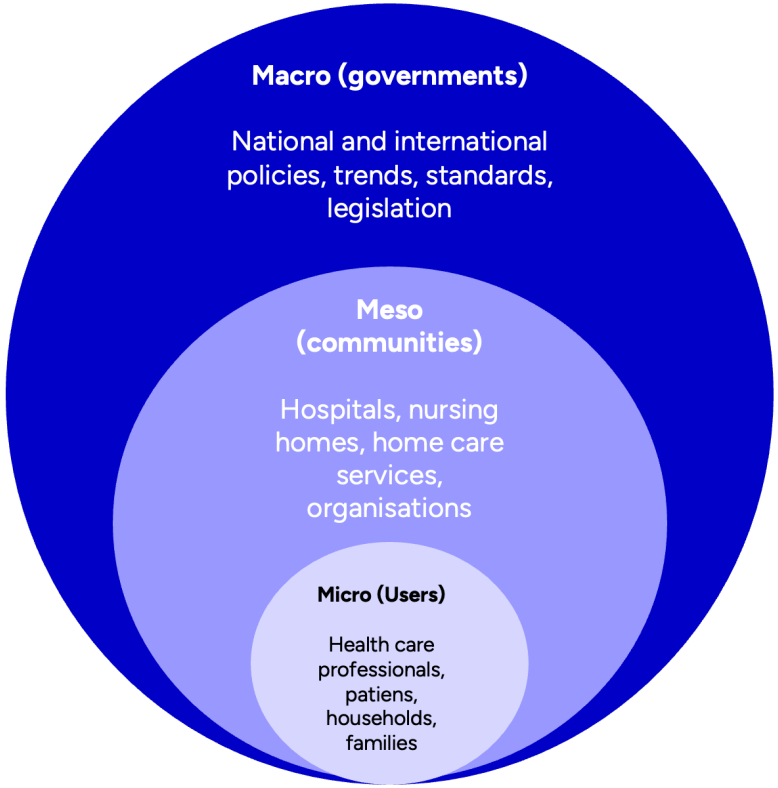
- Think about the design and development of these technologies
- Who are the users of these technologies?
- Why do users matter in the design?
- Can users improve the design? How?
- What kind of values are embedded in these technologies?

Break 15 min!



Social science methods in engineering fields

- Social Science methods are systematic tools to collect and analyse data to understand users in different contexts
- Quantitative methods to explore frequences, prevalences and associations in technology use
- Qualitative methods to understand *why* and *how* users adopt or reject certain technologies, what kind of meanings they associate with it



Methodology in this course

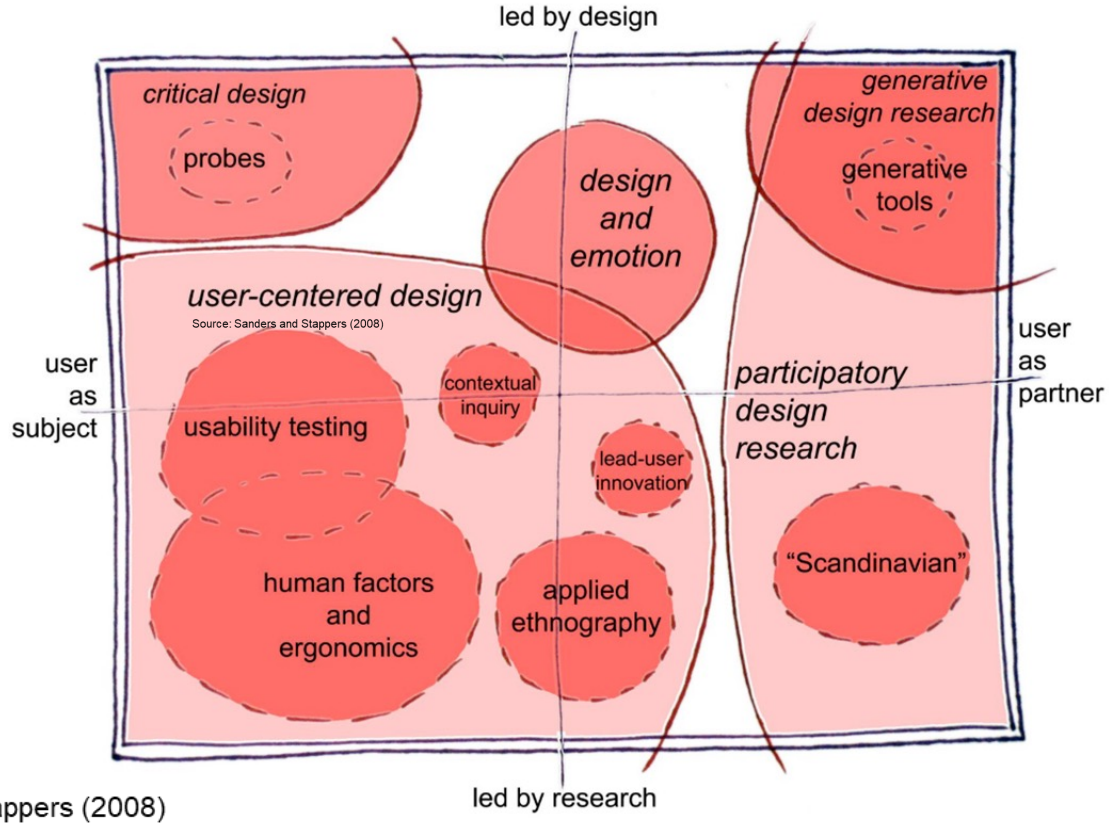
Quantitative methods

- Quantitative methods in digital media and consumer research
- Principles of quantitative research
- Designing survey questionnaire
- Statistical procedures
- Usability scales and questionnaires
- Free online tools

Qualitative methods

- Principles of qualitative research
- Social constructionism
- Phenomenology
- Digital and design ethnography
- Using ethnography in design process
- Interviewing
- Thematic analysis

User Involvement: Relationship among approaches



Source: Sanders and Stappers (2008)



Examples of user studies

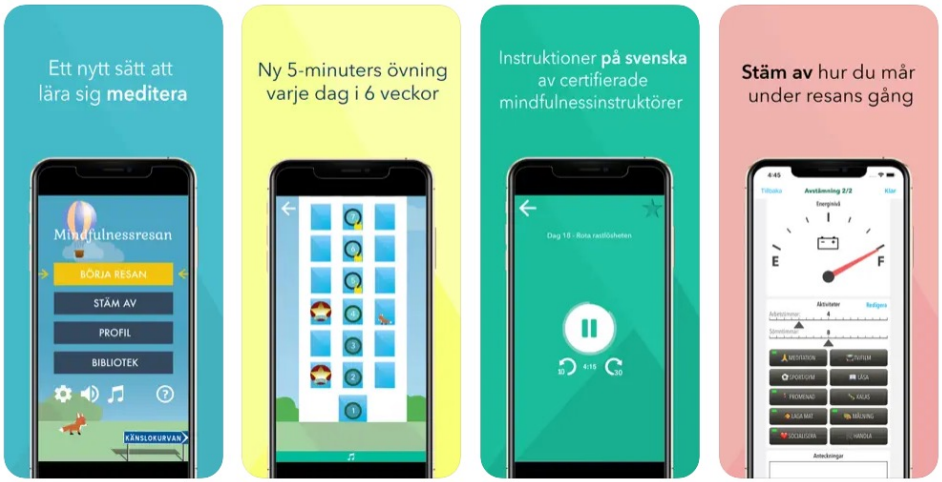
Mindfulness-based interventions for mental well-being

- Mindfulness means ‘paying attention in a particular way, on purpose, in the present moment, and non-judgmentally’ (Kabat-Zinn, 1994)
- Mindfulness-based interventions are used in health care to prevent and treat chronic pain, sleep disorders, anxiety and depression
- Swedish Healthcare Guide (1177) defines mindfulness as a method which involves focusing on the present with the help of breathing which can train the ability for conscious presence



Digital Mindfulness

- Mindfulness apps have increased the availability of MBIs.
- Voice-based guided mindfulness exercises, self-tracking of mindfulness activities, setting personal goals and receiving reminder notifications
- Self-guided format, where users do not have a personal interaction with a mindfulness instructor
- Influencing user's emotional state through increasing reflexivity and emotional awareness.



Method

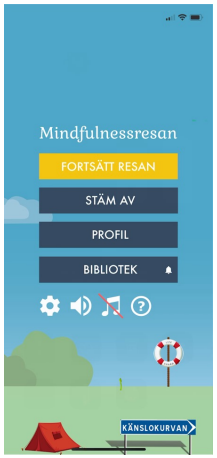
Part 1

User study: **non-conversational** mindfulness app

Pre-usage study

1. WHO-5 index
2. 5 Facet
3. Perceived stress

Previous use of mHealth apps



Post-usage study

1. WHO-5 index
2. 5 Facet
3. Perceived stress

mAUQ

Semi-structured interview

Part 2

Participatory design of **conversational** mindfulness app

Workshop 1

Co-design a **dialogue model** for a conversational application

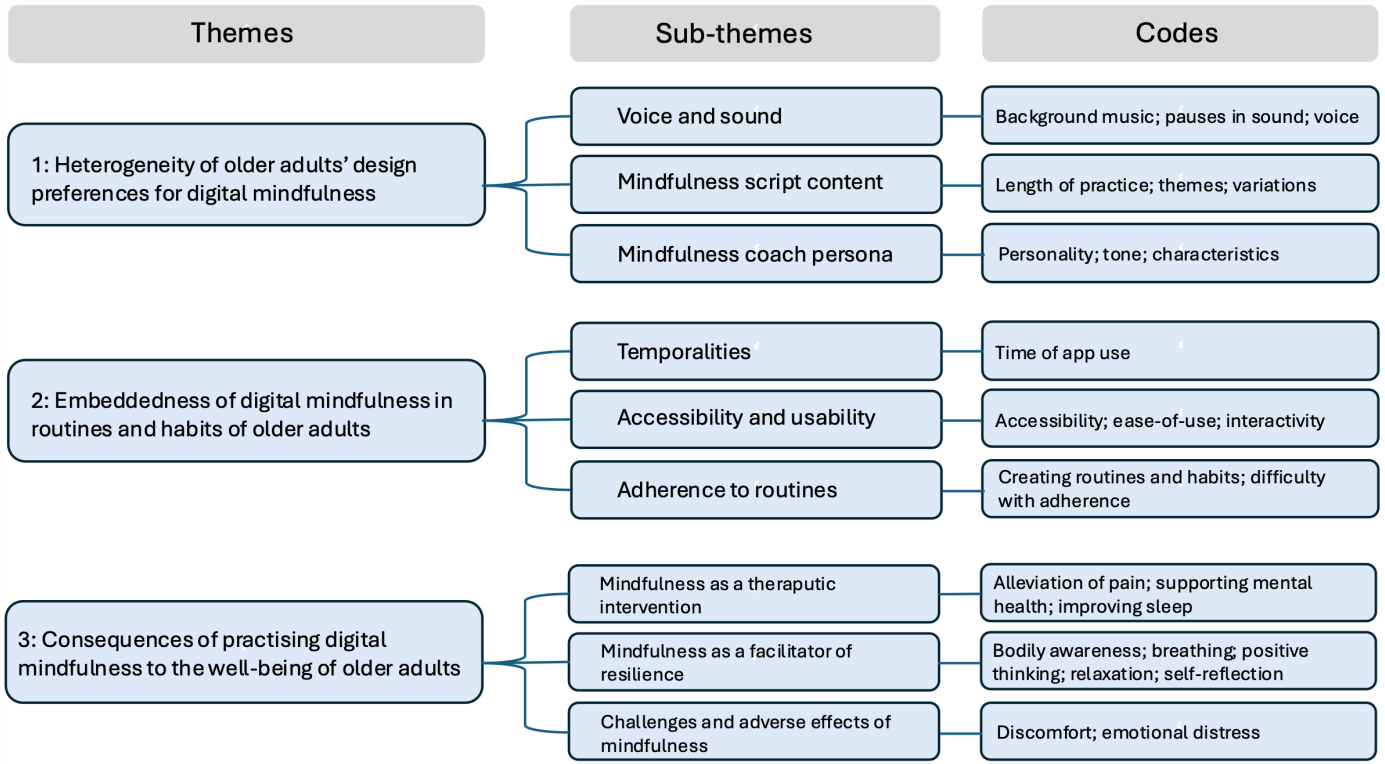
Mindfulness style
Persona and communication
Conversational flow
Pacing of interactions

Workshop 2

Co-design a **user interface** for a conversational application

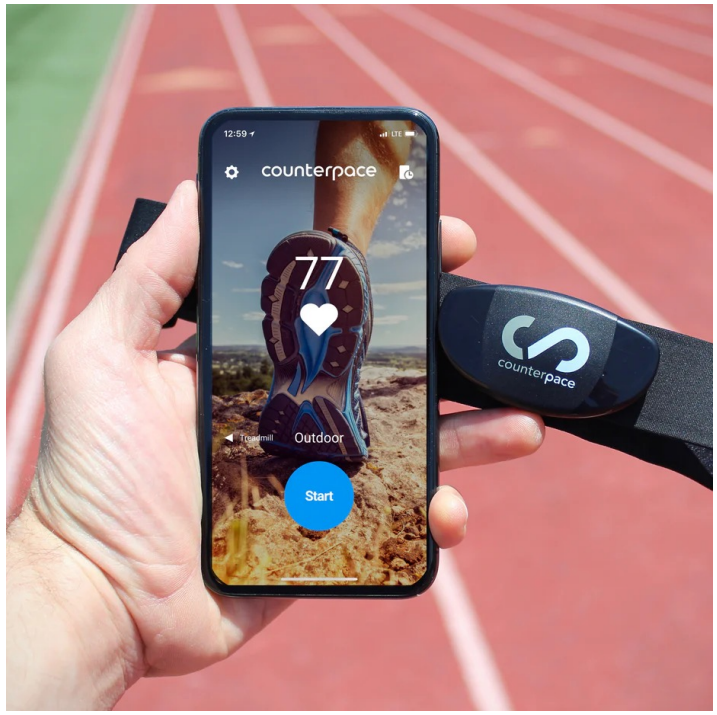
Human-like attributes
Avatar
Auditory interaction
Colours and visual experience
Navigation and layout

Thematic analysis of interview data



Wearables with real-time feedback (WRF) for physical activity

- WRF enable feedback to the user while the activity is on-going
- Auditory feedback (sound, music, audio)
- Wearables as self-care technologies vs. medical technologies
- User engagement still remains a challenge, particularly among vulnerable groups



User perceptions of wearables with real-time feedback

Use of wearables without real-time feedback

- Maintenance of embodied autonomy and resistance to performance tracking
- Contextual and temporal meaning-making
- Tensions and ambivalent emotional responses

Wearables with real-time feedback

- Disrupted natural flow
- Learning and gradual integration of device-guided syncronisation
- Raising bodily awareness through real-time multisensory feedback

Lu, T-Y., Rosato, A., Dual, S., Kuoppamäki, S: (2025). Lived Experiences of Older Adults Using Wearables with Real-Time FeedbackStudy. JMIR mHealth and uHealth

User perceptions of wearables with real-time feedback

Wearables with real-time feedback

- Relational meaning-making and interpretation of health data
- Technological barriers hindering PA enhancement

Wearables with real-time feedback

- Disruption and cognitive overload (limited device use)
- Bodily awareness (moderate device use)
- Integration and internalisation (extended device use)

Lu, T-Y., Rosato, A., Dual, S., Kuoppamäki, S: (2025). Lived Experiences of Older Adults Using Wearables with Real-Time FeedbackStudy. JMIR mHealth and uHealth

AI chatbot for positive body image

- **Nutrition technologies** today are designed to monitor, collect and process data on dietary intake to optimise nutrition choices
- **Body image satisfaction,** defined as person's beliefs, thoughts and feelings towards one's body can have a significant role in maintaining healthy lifestyles and prevention of disordered eating behaviours

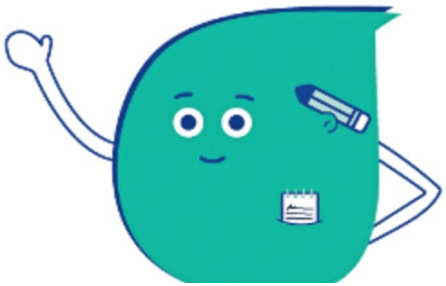
1) How can a LLM-based chatbot be designed and used to enhance body image satisfaction in young adults?
2) What are the challenges and benefits of using a dialogue model to raise awareness of body image satisfaction?
3) How can trust be achieved in LLM-based interactions in promoting positive body image for young adults?





Eating disorders in Human-Computer interaction

- Positive Body Image chatbot that provides education and coping skills for targeting body image concerns (Beilhardz, 2021)
- Prevention chatbot for eating disorders (Chan et al., 2022)
- Mindful eating (Khot et al., 2022, Zhang et al., 2023)
- Eating disorder applications to monitor habits, thought and feelings in eating (Devakumar et al., 2021)
- Social media and eating disorders (Pater et al., 2021)
- Usage of weight loss applications (Eikey et al., 2017)

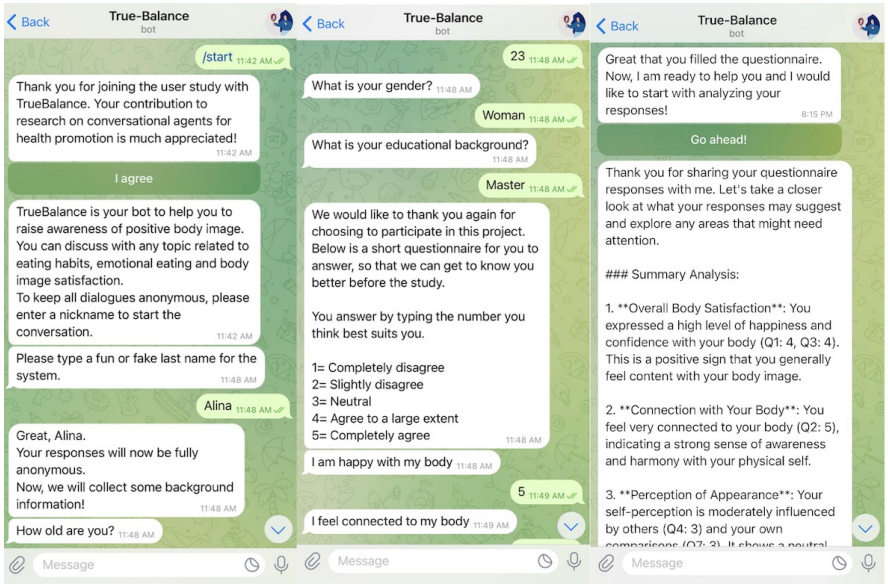


There are a range of signs to look out for if you're worried. These signs might include, but are not limited to:
- ☆ Restrictive or irregular eating e.g. skipping meals, fasting, avoiding particular food groups
- ☆ Feeling preoccupied or constantly thinking about food or body
- ☆ Binge eating or feeling out of control with food
- ☆ Compensatory behaviours e.g. vomiting, laxatives, diuretics, chewing and spitting
- ☆ Excessive exercise, including when sick or injured
- ☆ Difficulty changing routines, particularly around food or exercise

Beilharz, F., Sukunesan, S., Rossell, S. L., Kulkarni, J., & Sharp, G. (2021). Development of a Positive Body Image Chatbot (KIT) With Young People and Parents/Carers: Qualitative Focus Group Study. Journal of Medical Internet Research, 23(6), e27807. https://doi.org/10.2196/27807

TrueBalance Chatbot (TBC)

- TrueBalance 0.1 was developed through OpenAI's GPT Builder (Generative Pre-trained Transformer (GPT 4) model) through prompt engineering and fine-tuning the model
- Biomedical determinants of Eating Disorders (genetic, neurobiological and hormonal factors) as **background information**
- Principles of Cognitive Behavioral Therapy (recognition of negative thoughts or emotions) as **conversational style**
- TrueBalance 0.2 was developed with OpenAI's assistant API to enable anonymous conversations, memory of previous discussions and user-friendly interface



Data and methods

- User study with young adults aged 20-30 was conducted in September-November 2024 (n=15)
- Participants were asked to use TrueBalance Chatbot (TBC) for 1-week period in their homes at least 3 times during 7 days
- Interviews and questionnaires ***before and after*** using TB
- Survey questionnaires about Body Image Satisfaction and Social Media Use (n=15)
- Semi-structured interviews about the user experience, perceived benefits and challenges (n=15)



Figure 1. Homepage of the TrueBalanceApp showing core features



Figure 2. Chat-interface of the TrueBalanceApp

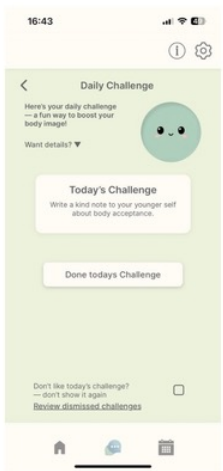


Figure 3. The Daily Challenge interfacace of the TrueBalanceApp

Maintaining body image awareness without the AI-agent

Theme	Sub-theme
Maintaining and challenging positive body image	Self-acceptance and self-appreciation
	Valuing body functionality
	Body image insecurities
	Influence of cultural and societal standards
	Paradoxical influence of social media
Strategies for improving body positivity	Social support networks
	Exercise and physical well-being
	Positive self-talk and self-acceptance
	Make-up and cosmetics
	Social media detox

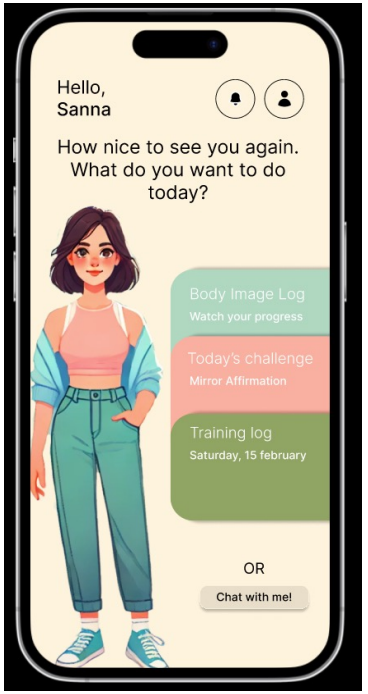
Advantages and disadvantages of using an AI-agent

Theme	Category
Facilitating body image awareness and self-reflection	Increased knowledge and awareness
	Facilitating reflective thinking
	Promoting self-acceptance and emotional resilience
Availability of conversational support	Instant and constant availability
	Private and non-judgemental communication
	Emotion-sensitive conversations
Discontinuities in the user engagement	Overwhelming messages
	Desire for continuing the conversation declined over time
	Challenges in turning advice into action
	Issues with the chatbot's responsiveness

Advantages and disadvantages of using an AI-agent for positive body image

- Increased understanding of eating disorders and emotional eating
- Medium for affirming pre-knowledge, instead of changing body image
- Facilitating reflective thinking, increased awareness cultivating positive thinking patterns
- Promoting self-acceptance and emotiona resilience

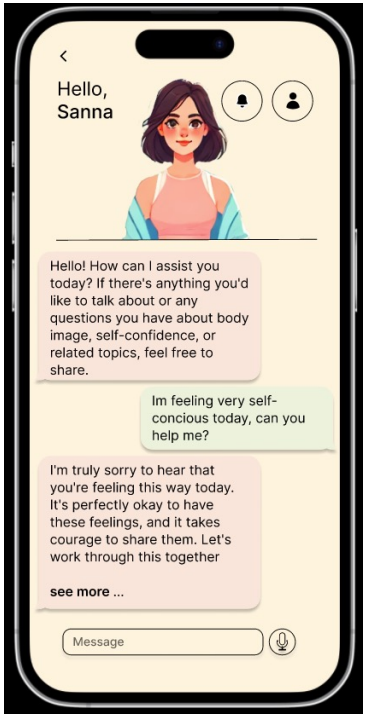
I learned a lot from the app. It was good to know what is healthy, and I would learn a couple of facts here and there about how to eat more balanced, especially when I'm not feeling hungry or if I'm feeling depressed. (P1, 24, female)



Availablility of conversational support

- Instant support when other resources are not available
- Sense of comfort for being available and accessible all the time
- Private and non-judgemental conversations provide a safe conversational environment
- However, some messages were considered overwhelming and interest towards maintaining the conversations decreased over time

(The agent) can give instant support. If I have a thought right now, my doctor might not be available right now to give me some information on what I need or make me calm down. (P12, 25, female)



Group discussion for final assignment

Discuss ideas for development of the user study in small groups

- What kind of technologies are you interested to develop?
- What kind of projects you have worked previously?
- What kind of (user) methods you have used?
- What would be an interesting technology for the user study?



Preparations for next week (Seminar 25/3)

- Purpose is to present and discuss the current stage of the user study, give and receive feedback to other's work and raise any questions in different stages of the user study
- You need to be prepared to present the *preliminary idea* for your user study (technology, product, context, user group)
- You can work individually, pairs or in small groups
- Please use power points in presentation!



→ **Group formation for final assignment**